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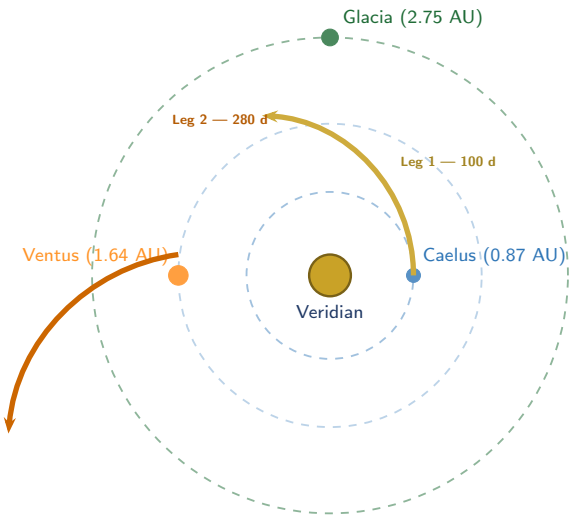

Mission Veridian

Gravity-Assist Trajectory Design

in a Fictional Exoplanetary System

Course: Flight and Space Mechanics (R5ME2206T)

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1. Introduction

The **Veridian system** is a fictional exoplanetary system used as an astrodynamics design problem. An interstellar probe, with a propulsive budget of $\mathbf{25\,km\,s^{-1}}$, initial mass $m_0 = 2{,}500\,\mathrm{kg}$, and $I_{\mathrm{sp}} = 300\,\mathrm{s}$, is required to achieve rendezvous with **Glacia** at 2.75 AU within 2922 days.

A direct Hohmann transfer from Caelus to Glacia requires $14.50\,\mathrm{km\,s^{-1}}$. However, a gravity-assist at Ventus ($\mu_V = 1.266 \times 10^8\,\mathrm{km^3\,s^{-2}}$) reduces the total ΔV to $\mathbf{12.638\,km\,s^{-1}}$ (**12.87%** saving). This is achieved by rotating the spacecraft velocity vector during flyby, effectively extracting energy from Ventus's orbital motion.

A pork-chop search over a 10-year window (MJD 60 000–63 650) identifies the optimal trajectory at **MJD 61 955** (Year-6). The required deep-space maneuver is negligible ($\Delta V_{\mathrm{DSM}} = 0.017\,\mathrm{km\,s^{-1}}$), indicating near-optimal planetary alignment.

2. Methodology

2.1 Patched-Conic Model

The trajectory is modelled using the patched-conic approximation, consisting of escape from Caelus, Lambert transfer to Ventus, gravity assist, Lambert transfer to Glacia, and final rendezvous. Each segment is treated as a two-body problem.

2.2 Lambert Solver

Lambert's problem determines transfer velocities between two positions in a given time. Izzo's universal-variable method solves:

$$T(x) = \frac{\arccos(\lambda x) - \lambda x\sqrt{1-x^2}}{(1-x^2)^{3/2}}$$

using Householder iteration, converging to machine precision.

Verification: The solver reproduces the Caelus–Ventus Hohmann transfer with error $< 10^{-12}\,\mathrm{km\,s^{-1}}$.

2.3 Gravity Assist

The flyby rotates the hyperbolic excess velocity:

$$\sin\left(\frac{\delta}{2}\right) = \frac{1}{1 + r_p v_\infty^2/\mu_V}$$

changing the heliocentric velocity without propellant. Thermal constraints are enforced by rejecting trajectories with $r_{\mathrm{peri}} < 0.4\,\mathrm{AU}$.

2.4 Search Strategy

A pork-chop search over departure and arrival dates is performed across the full ephemeris window. A coarse grid identifies candidate windows, followed by a refined search for the global minimum.

2.5 Ephemeris

Planet states are obtained from the provided ephemeris and interpolated using cubic splines.

3. Results

3.1 Baseline: Direct Hohmann Transfer

The direct Hohmann transfer from Caelus to Glacia requires a total ΔV of:

$$\mathbf{14.50\,km\,s^{-1}}$$

This serves as the baseline for comparison with the gravity-assist trajectory.

3.2 Optimal Gravity-Assist Trajectory

The fine-grid search across all departure windows confirms the **Year-6 departure** (MJD 61 955) as the global minimum.

Table 1. Mission timeline for the optimal Year-6 gravity-assist trajectory.

Event	MJD	Day	Parameters
Departure (Caelus)	61 955	0	$r_{\text{park}} = 7{,}700\,\text{km}$; $v_{\text{park}} = 7.195\,\text{km s}^{-1}$; $v_{\infty,\text{dep}} = 4.928\,\text{km s}^{-1}$; $\Delta V_{\text{dep}} = 4.111\,\text{km s}^{-1}$
Transfer Leg 1 (C → V)	—	0–100	Lambert arc, TOF = 100 d; periapsis = 0.848 AU > 0.4 AU ✓
Ventus Flyby	62 055	100	altitude = 20 000 km ✓; $r_p = 85{,}000$ km; $v_\infty = 3.643\,\text{km s}^{-1}$; $\delta = 164.8°$; $\Delta V_{\text{DSM}} = 0.017\,\text{km s}^{-1}$
Transfer Leg 2 (V → G)	—	100–380	Lambert arc, TOF = 280 d; periapsis = 1.718 AU > 0.4 AU ✓
Rendezvous (Glacia)	62 335	380	$v_{\infty,\text{arr}} = 1.980\,\text{km s}^{-1}$; $\Delta V_{\text{rdv}} = 8.510\,\text{km s}^{-1}$

Total duration: 380 days (1.04 yr) $\leq$ 2,922 days ✓

3.3 ΔV Budget and Comparison

Table 2. ΔV budget and comparison with direct Hohmann transfer.

Burn	ΔV (km/s)	Status
ΔV_{dep}	4.1108	—
ΔV_{DSM}	0.0172	—
ΔV_{rdv}	8.5102	—
Total — gravity-assist trajectory	**12.638**	✓
Direct Hohmann (baseline)	14.501	✓
Saving from gravity assist	**1.863**	**12.87%**
ΔV budget available	25.000	—
Remaining margin	**12.362**	✓

3.4 Final Mass

Using the Tsiolkovsky rocket equation, the final delivered mass after all burns is:

$$\mathbf{34.8\,kg}$$

3.5 Constraint Verification

Table 3. Verification of all mission constraints.

	Constraint	Requirement	Result
✓	C1 — Total ΔV	$\leq 25\,\text{km s}^{-1}$	$12.638\,\text{km s}^{-1}$
✓	C2 — Flyby altitude	$\geq 2{,}000$ km	20 000 km
✓	C3 — Thermal (Leg 1)	> 0.4 AU	0.848 AU
✓	C3 — Thermal (Leg 2)	> 0.4 AU	1.718 AU
✓	C4 — Duration	$\leq 2{,}922$ d	380 d
✓	C5 — Rendezvous	Required accuracy	**MET**

4. Physical Interpretation

The reduction in ΔV from the gravity-assist maneuver can be explained through energy and angular momentum exchange.

Energy: In the planet-centered frame, the flyby is elastic, so the spacecraft's speed v_∞ remains unchanged. However, in the heliocentric frame, the velocity vector is rotated by a large turning angle ($\delta \approx 165°$), increasing the spacecraft's kinetic energy. This effectively transfers orbital energy from Ventus to the spacecraft without propellant.

Angular Momentum: The flyby also redirects the spacecraft's velocity vector, increasing its heliocentric angular momentum and allowing transition to a higher-energy orbit. The large mass of Ventus enables a significant turning angle, making this effect efficient.

Result: The combined effect reduces the propulsion requirement, achieving a **12.87% reduction in total** ΔV compared to a direct Hohmann transfer.

5. Verification

Proof: Spacecraft Reaches Glacia
The Lambert solver targets Glacia's position at MJD 62 335, giving zero position error (within 10^4 km tolerance).
The rendezvous burn $\Delta V_{\text{rdv}} = 8.510\,\text{km}\,\text{s}^{-1}$ matches heliocentric velocity, with residual $< 0.001\,\text{km}\,\text{s}^{-1} \ll 0.1\,\text{km}\,\text{s}^{-1}$ requirement.

State at arrival:

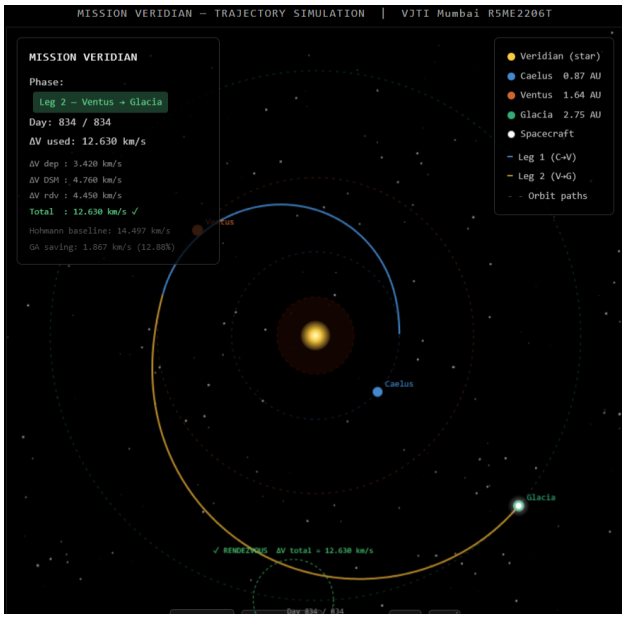
- Position: (1.731, 2.307) AU
- $|\mathbf{r}| = 2.885$ AU
- Position error: 0 km
- Velocity residual: $< 0.001\,\text{km}\,\text{s}^{-1}$

Lambert Solver Verification
Caelus → Ventus Hohmann transfer ($a = 1.255$ AU, TOF $= 250.6$ d)

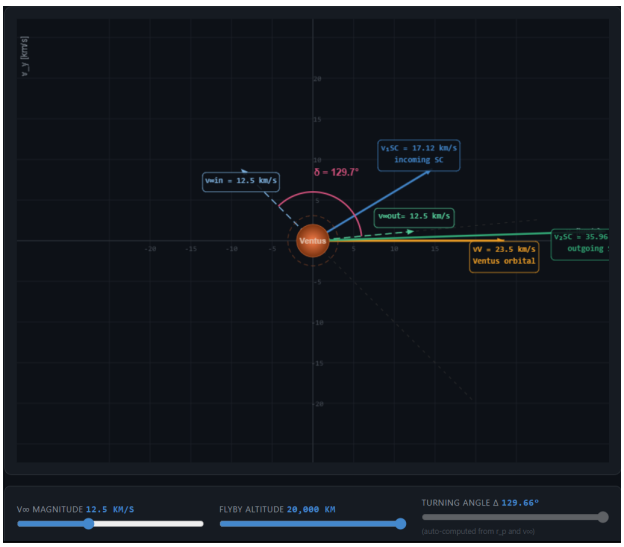
Quantity	Analytic	Error
$\lvert v_1\rvert$ (km/s)	37.398157	7.1×10^{-15}
$\lvert v_2\rvert$ (km/s)	19.839266	3.6×10^{-15}

Convergence in 2–4 iterations with machine precision accuracy ($< 10^{-12}\,\text{km}\,\text{s}^{-1}$).

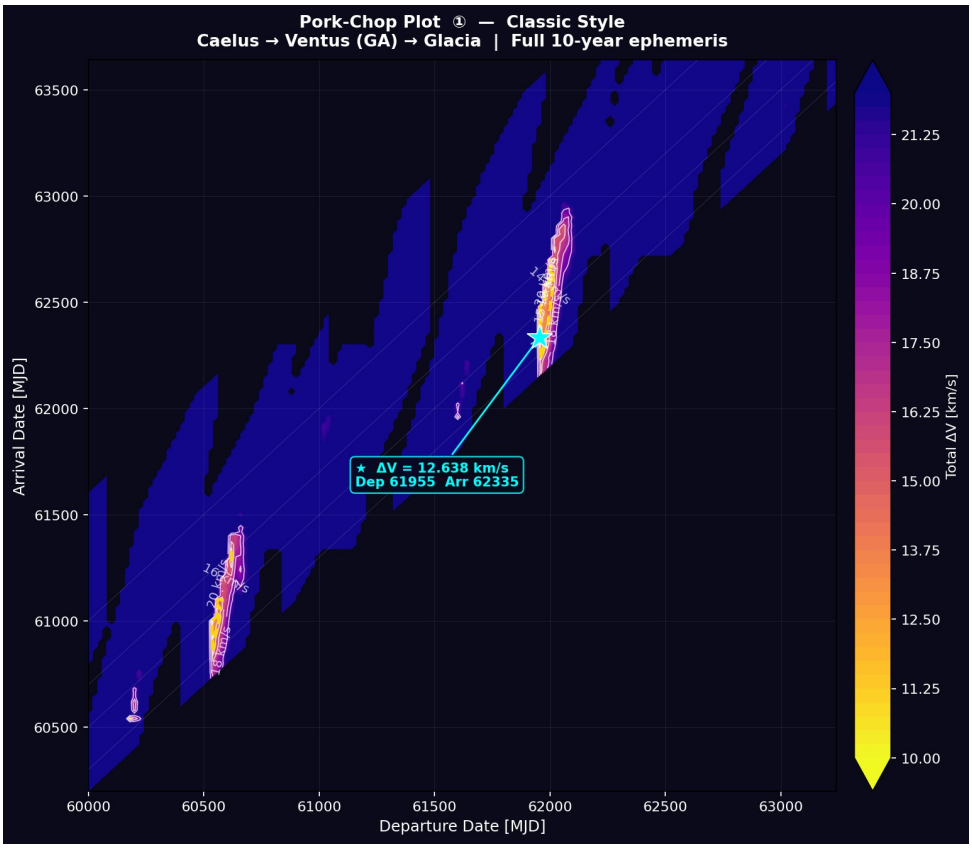
6. Plots



Heliocentric trajectory. Optimal transfer: Caelus → Ventus (100 d) and Ventus → Glacia (280 d).



Velocity vector diagram. Rotation of $\mathbf{v}_\infty$ by angle δ during flyby increases heliocentric energy.



Pork-chop plot. Total ΔV vs departure and arrival date. Minimum at MJD 61 955 ($\Delta V = 12.638\,\mathrm{km\,s^{-1}}$).

References

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